# Analysis of Addition Modulo $2^n$ on Boolean Function

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***Abstract*—In order to analyze the impact on the security of cryptographic algorithm produced by the linear approximations of addition modulo $2^n$ with XOR, this paper firstly translates the operation of addition modulo $2^n$ into vector Boolean function with dimension of *n*. The component functions can be obtained through a recurrence formula which does not require a recall of the carry function. Then we explore the cryptographic properties of *n*-dimensional Boolean functions of addition modulo $2^n$, and the results indicate that the component functions are all *k*th-order quasi-Bent functions. The *n*-dimensional vector has the first order-correlation immunity, rather than the second-order correlation immunity.**

***Keywords-XOR; addition modulo$2^n$; Boolean function; kth-order quasi-Bent functions; Correlation Immunity;***

## I. INTRODUCTION

Addition modulo $2^n$ and XOR, which are based on the ring $Z_{2^n}$ and the field $Z_2$ respectively, are two different operations. Compared with XOR, addition modulo $2^n$ has much better nonlinearity. Therefore, cryptographic algorithm designers often use them together to enhance the security of cryptographic algorithms. This encryption is widely used like e.g. in the block ciphers Safer [1]and Mars [2], in the stream ciphers Snow [3], and in the hash algorithm MD5 [4],SHA-1 [5] and the SHA-3 candidates BLAKE [6] etc.

To analyze addition modulo $2^n$ and XOR which are contained in the cryptographic algorithm, we usually translate one to the other so that the analysis of two different operations can be translated into the analysis of a single operation. The consistent degree of addition modulo $2^n$ and XOR has been discussed in [7, 8]. Hiroshi and Alexis[9,10] present the differential probability of addition modulo $2^n$.

In this paper, the operation of addition modulo $2^n$ is viewed as a vector Boolean function, and then we deduce a recurrence formula that does not require a recall of the carry function about $X+Y$ mod $2^n$. Lastly, we study the cryptographic properties of addition modulo $2^n$.

## II. PRELIMINARIES

In this section, we will introduce the notion to be used later. Suppose that $Z_{2^n}$ is a ring, and $Z_2$ is a field with two elements 0 and 1. we know according to the knowledge of algebra that for any element $X \in Z_{2^n}$, $X$ can be represented in the form of $X=\sum_{i=0}^{n-1} 2^i x_i$, which we denote as $X=(x_{n-1}, \cdots, x_1, x_0)$, where $x_i \in \{0,1\}$, $0 \le i \le n-1$. For any two elements $X=(x_{n-1}, \cdots, x_1, x_0)$ and $Y=(y_{n-1}, \cdots, y_1, y_0)$, $X+Y$ mod $2^n$ is still an element over ring $Z_{2^n}$. Therefore $X+Y$ mod $2^n$ can also be represented in the form of

$$X+Y=\sum_{i=0}^{n-1} 2^i f_i \triangleq (f_{n-1}, \cdots, f_1, f_0), \qquad (1)$$

where $f_i$ are decided by $(x_{n-1}, \cdots, x_1, x_0)$ and $(y_{n-1}, \cdots, y_1, y_0)$, and are all Boolean functions with $2n$ variables.

The aim of the paper is to get the relationship between $f_i$, $(x_{n-1}, \cdots, x_1, x_0)$ ,and $(y_{n-1}, \cdots, y_1, y_0)$, then to explore the cryptographic properties of $f_i$.

In the following of the paper, the symbol "+" always means addition modulo $2^n$, and "$\oplus$" means exclusive OR, that is XOR.

## III. THE EXPRESSION OF ADDITION MODULE $2^N$ ON VECTOR BOOLEAN FUNCTION

Paper [11] has ever studied the structure of $X+Y$, and translated it into a *n*-dimensional Boolean function with 2n variables as follows:

**Theorem 1**[11] Let $X, Y \in Z_{2^n}$, and $f(X,Y)=X+Y$ mod $2^n$ can be represented as $(f_{n-1}, \cdots, f_1, f_0)$. Then

$$\begin{cases} f_0 = x_0 \oplus y_0 \oplus c_0, c_0 = 0 \\ f_i = x_i \oplus y_i \oplus c_i, c_i = x_{i-1}y_{i-1} \oplus x_{i-1}c_{i-1} \oplus y_{i-1}c_{i-1}, \forall 1 \le i \le n-1 \end{cases}$$

According to theorem 1, we have

$$f_0 = x_0 \oplus y_0$$
$$f_1 = x_1 \oplus y_1 \oplus x_0 y_0$$
$$f_2 = x_2 \oplus y_2 \oplus x_1 y_1 \oplus x_0 y_0 y_1 \oplus x_1 x_0 y_0$$
$$f_3 = x_3 \oplus y_3 \oplus x_0 x_1 x_2 y_0 \oplus x_1 x_2 y_1 \oplus x_0 x_2 y_0 y_1 \oplus x_2 y_2 \oplus x_0 x_1 y_0 y_2 \oplus x_1 y_1 y_2 \oplus x_0 y_0 y_1 y_2$$

The number of items in the component functions $f_i$ increases with $i$ by a geometric index, and the computational complexity of the above algorithm is $O(2i2^{2i})$[12]. Meanwhile, the computation algorithm presented in theorem 1 need a large space to store the carry functions $c_i$, which will decrease its efficiency. Now we give a recurrence formula which does not require the carry function.

**Theorem 2** Let $X, Y \in Z_{2^n}$, and $f(X,Y)=X+Y$ mod $2^n$ can be represented as $(f_{n-1}, \cdots, f_1, f_0)$. Then

$$\begin{cases} f_0 = x_0 \oplus y_0, i = 0 \\ f_i = x_i \oplus y_i \oplus x_{i-1}y_{i-1} \oplus (x_{i-1} \oplus y_{i-1})(f_{i-1} \oplus x_{i-1} \oplus y_{i-1}), \forall 1 \le i \le n-1 \end{cases}$$

*Proof*: Based on Theorem 1, it is easy to conclude that

$$f_i = x_i \oplus y_i \oplus c_i, c_i = f_i \oplus x_i \oplus y_i$$
$$f_{i+1} = x_{i+1} \oplus y_{i+1} \oplus c_{i+1}$$

978-0-7695-4559-2/11 $26.00 © 2011 IEEE
DOI 10.1109/MINES.2011.131

$= x_{i+1} \oplus y_{i+1} \oplus x_i y_i \oplus x_i c_i \oplus y_i c_i$
$= x_{i+1} \oplus y_{i+1} \oplus x_i y_i \oplus (x_i \oplus y_i) c_i$

By replacing $c_i$ with $f_i \oplus x_i \oplus y_i$, we can deduce that $f_{i+1} = x_{i+1} \oplus y_{i+1} \oplus x_i y_i \oplus (x_i \oplus y_i)(f_i \oplus x_i \oplus y_i)$.

The recurrence formula in theorem 2 eliminates the need of $c_i$ for storage and recall. Thus it can improve the algorithm speed. Meanwhile, it is significant for us to analyze the cryptographic properties of addition modulo $2^n$.

## IV. CRYPTOGRAPHIC PROPERTIES OF ADDITION MODULE $2^N$

The cryptographic properties of Boolean function can be studied by its Walsh spectra.

**Definition 1[13]** Let *f(x)*, $x \in Z_2^n$ be a Boolean function with n variables. Then

$$S_{(f)}(w) = \frac{1}{2^n} \sum_{x \in Z_2^n} (-1)^{f(x)+w \cdot x}, w \in Z_2^n$$

is said to be the Walsh cyclic spectra of *f(x)*.

The functions with three-valued Walsh cyclic spectra, and include Bent functions as its proper subset can have significant cryptographic properties.

**Definition 2[13]** Let *f(x)*, $x \in Z_2^n$ be a Boolean function with *n* variables. *f(x)* is said to be a *k*th-order quasi-Bent function if $S_{(f)}(w) = 0$ *or* $\frac{1}{2^{n-k}}, w \in Z_2^n$.

**Theorem 3.** Let $X, Y \in Z_{2^n}$, and $f(X,Y) = X+Y \bmod 2^n$ can be represented as $(f_{n-1}, \cdots, f_1, f_0)$. Then $f_0 = x_0 \oplus y_0$ is a linear function, and $f_i (1 \le i \le n-1)$ is $(2n\text{-}2i)$ order quasi-Bent function.

Proof: We prove theorem by induction.

**Step 1:** For $i = 1$, $f_1 = x_1 \oplus y_1 \oplus x_0 y_0$.

$$S_{(f_1)}(w) = 2p\{f_1 \oplus w_1 \cdot (X,Y) = 0\} - 1.$$

When $w_1 = (0, \cdots, 0, 1, 0; 0, \cdots, 0, 1, 0)$,

$$S_{(f_1)}(w_1) = 2P\{x_1 \oplus y_1 \oplus x_0 y_0 \oplus x_1 \oplus y_1 = 0\} - 1$$
$$= 2P\{x_0 y_0 = 0\} - 1$$
$$= \frac{1}{2};$$

When $w_1 = (0, \cdots, 0, 1, 0; 0, \cdots, 0, 1, 1)$,

$$S_{(f_1)}(w_1) = 2P\{(x_0 \oplus 1) y_0 = 0\} - 1 = \frac{1}{2};$$

When $w_1 = (0, \cdots, 0, 1, 1; 0, \cdots, 0, 1, 0)$,

$$S_{(f_1)}(w_1) = 2P\{x_0 (y_0 \oplus 1) = 0\} - 1 = \frac{1}{2};$$

When $w_1 = (0, \cdots, 0, 1, 1; 0, \cdots, 0, 1, 1)$,

$$S_{(f_1)}(w_1) = 2P\{x_0 y_0 \oplus x_0 \oplus y_0 = 0\} - 1 = -\frac{1}{2}.$$

For the other case of $w_1$, $S_{(f_1)}(w_1) = 0$.

The results show that for any $w \in Z_2^{2n}$, $S^2{}_{(f_1)}(w_1) = 0$ or $\frac{1}{2^2}$. So $f_1$ is a $(2n\text{-}2)$-order quasi-Bent function.

**Step 2:** Suppose the theorem is true for $i$ ($0 \le i \le n-2$), i.e. $f_i$ is a $(2n\text{-}2i)$-order quasi-Bent function. Hence when

$$\begin{aligned} w_i &= (*, \cdots, *, 1, 0, \cdots, 0; *, \cdots, *, 1, 0, \cdots, 0;), \\ w_i' &= (*, \cdots, *, 0, 0, \cdots, 0; *, \cdots, *, 0, 0, \cdots, 0;) \end{aligned} \quad (2)$$

where the former *i* bits, and the elements from the (*n*+1)th component to the (*n*+*i*)th component of $w_i, w_i'$ are all unknown quantity. Then we have

$$S_{(f_i)}(w_i) = 2P\{f_i \oplus w_i \cdot (X,Y) = 0\} - 1$$
$$= 2P\left\{f_i \oplus x_i \oplus y_i = w_i' \cdot (X,Y)\right\} - 1$$
$$= \pm \frac{1}{2^i},$$

and $P\{f_i \oplus x_i \oplus y_i = w_i' \cdot (X,Y)\} = \frac{1}{2} \pm \frac{1}{2^{i-1}}$.

For any other $w \in Z_2^{2n}$ which is different with that in formula (2), we can get $S_{(f_i)}(w) = 0$

**Step 3:** For $i$+1, according to theorem 2, we have

$$f_{i+1} = x_{i+1} \oplus y_{i+1} \oplus x_i y_i \oplus (x_i \oplus y_i)(f_i \oplus x_i \oplus y_i),$$

When

$$\begin{aligned} w_{i+1} &= (*, \cdots, *, 1, 0, \cdots, 0; *, \cdots, *, 1, 0, \cdots, 0;), \\ w_{i+1}' &= (*, \cdots, *, 0, 0, \cdots, 0; *, \cdots, *, 0, 0, \cdots, 0;) \end{aligned} \quad (3)$$

Where the former *i*+1bits, and the elements from the (*n*+1)th component to the (*n*+*i*+1)th component of $w_i, w_i'$ are all unknown quantity.

$$S_{(f_{i+1})}(w_{i+1}) = 2P\{f_{i+1} \oplus w_{i+1} \cdot (X,Y) = 0\} - 1$$
$$= 2P\left\{f_{i+1} \oplus x_{i+1} \oplus y_{i+1} = w_{i+1}' \cdot (X,Y)\right\} - 1$$
$$= 2P\left\{x_i y_i \oplus (x_i \oplus y_i)(f_i \oplus x_i \oplus y_i) = w_{i+1}' \cdot (X,Y)\right\} - 1$$

When the (*i*+1)th component and the (*n*+*i*+1)th component of $w_{i+1}'$ are zero.

$$P\{f_{i+1} \oplus w_{i+1} \cdot (X,Y) = 0\}$$
$$= P\{x_i y_i \oplus (x_i \oplus y_i)(f_i \oplus x_i \oplus y_i) = w_i' \cdot (X,Y)\}$$
$$= \begin{cases} P\{w_i' \cdot (X,Y) = 0\} & x_i = 0, y_i = 0; \\ P\{f_i \oplus w_i' \cdot (X,Y) = 1\} & x_i = 1, y_i = 0; \\ P\{f_i \oplus w_i' \cdot (X,Y) = 1\} & x_i = 0, y_i = 1; \\ P\{w_i' \cdot (X,Y) = 1\} & x_i = 1, y_i = 1; \end{cases}$$

$$=\begin{cases}\frac{1}{2} & x_i=0,y_i=0;\\ \frac{1}{2}\pm\frac{1}{2^{i+1}} & x_i=1,y_i=0;\\ \frac{1}{2}\pm\frac{1}{2^{i+1}} & x_i=0,y_i=1;\\ \frac{1}{2} & x_i=1,y_i=1;\end{cases}$$

$$\begin{aligned}S_{(f_{i+1})}(w_{i+1})&=2P\{f_{i+1}\oplus w_{i+1}\cdot(X,Y)=0\}-1\\&=2\left[\frac{1}{4}\times\frac{1}{2}+\frac{1}{4}\times(\frac{1}{2}\pm\frac{1}{2^i})+\frac{1}{4}\times(\frac{1}{2}\pm\frac{1}{2^i})+\frac{1}{4}\times\frac{1}{2}\right]-1\\&=\pm\frac{1}{2^{i+1}}\end{aligned}$$

When the ($i$+1)th component of $w_{i+1}'$ is 1 and the ($n$+$i$+1)th component of $w_{i+1}'$ is zero.

$$\begin{aligned}&P\{f_{i+1}\oplus w_{i+1}\cdot(X,Y)=0\}\\&=P\{x_iy_i\oplus(x_i\oplus y_i)(f_i\oplus x_i\oplus y_i)=w_i'\cdot(X,Y)\oplus x_i\}\end{aligned}$$

$$=\begin{cases}P\{w_i'\cdot(X,Y)=1\} & x_i=0,y_i=0;\\ P\{f_i\oplus w_i'\cdot(X,Y)=0\} & x_i=1,y_i=0;\\ P\{f_i\oplus w_i'\cdot(X,Y)=1\} & x_i=0,y_i=1;\\ P\{w_i'\cdot(X,Y)=0\} & x_i=1,y_i=1;\end{cases}$$

$$=\begin{cases}\frac{1}{2} & x_i=0,y_i=0;\\ \frac{1}{2}\pm\frac{1}{2^{i+1}} & x_i=1,y_i=0;\\ \frac{1}{2}\pm\frac{1}{2^{i+1}} & x_i=0,y_i=1;\\ \frac{1}{2} & x_i=1,y_i=1;\end{cases}$$

$$\begin{aligned}S_{(f_{i+1})}(w_{i+1})&=2P\{f_{i+1}\oplus w_{i+1}\cdot(X,Y)=0\}-1\\&=2\left[\frac{1}{4}\times\frac{1}{2}+\frac{1}{4}\times(\frac{1}{2}\pm\frac{1}{2^i})+\frac{1}{4}\times(\frac{1}{2}\pm\frac{1}{2^i})+\frac{1}{4}\times\frac{1}{2}\right]-1\\&=\pm\frac{1}{2^{i+1}}\end{aligned}$$

By using the same method, we can get that when the ($i$+1)th component of $w_{i+1}'$ is zero, the ($n$+$i$+1)th component of $w_{i+1}'$ is 1, $S_{(f_{i+1})}(w_{i+1})=\pm\frac{1}{2^{i+1}}$ .When the ($i$+1)th component and the ($n$+$i$+1)th component of $w_{i+1}'$ are 1, $S_{(f_{i+1})}(w_{i+1})=\pm\frac{1}{2^{i+1}}$ .

For any other $w\in Z_2^{2n}$ which are different with that in formula (3), we can get $S_{(f_{i+1})}(w)=0$ .

Therefor for any $w\in Z_2^{2n}$, $S^2_{(f_{i+1})}(w)=0$ or $\frac{1}{2^{2(i+1)}}$ , which conclude the proof of theorem by definition 2.

From the above analysis, we obtain that $f_i$ is an ($2n$-$2i$)-order quasi-Bent function, which enable us to make a in-depth analysis on the other cryptographic properties of addition module $2^n$ , such as correlation immunity.

Paper [13] gives the linear combining lemma and spectrum criterion of correlation-immunity of vector Boolean function. We can study the correlation-immunity of addition modulo $2^n$ through a vector Boolean function over the field $Z_2$.

**Definition 3**[13] Let $f^{(k)}(x)=(f_1(x),\cdots,f_k(x))$ , $x\in Z_2^n$ be a vector Boolean function with $k$ dimensions. if $S_{(\alpha\cdot f^{(k)}(x))}(w)=0$ for any $w\in Z_2^n, 1\le W(w)\le m$ , $\alpha\in Z_2^k,\alpha\ne 0$ . $f^{(k)}(x)$ is said to be a m-order correlation-immune function.

**Theorem 4** Let $X,Y\in Z_{2^n}$ , and $f(X,Y)=X+Y$ mod $2^n$ can be represented as $(f_{n-1},\cdots,f_1,f_0)$. Then $(f_{n-1},\cdots,f_1,f_0)$ have the first-order correlation-immunity rather than the second-order correlation-immunity.

Proof: When $W(w)=1, w\in Z_2^{2n}$ ;

For any $\alpha\in Z_2^n$ ,

$$S_{(\alpha\cdot f)}(w)=2P\{\alpha\cdot f=w\cdot(X,Y)\}-1=0$$

So $f(X,Y)$ have the first-order correlation-immunity.

When $w=(0,\cdots,0,1,0;0,\cdots,0,1,0),\alpha=(0,\cdots0,1,,0)$ ;

$$\alpha\cdot f(X,Y)=(0,\cdots0,1,,0)\cdot(f_{n-1},\cdots,f_1,f_0)=f_1$$

$$\begin{aligned}S_{(\alpha\cdot f)}(w)&=2P\{\alpha\cdot f=w\cdot(X,Y)\}-1\\&=2P\{x_1\oplus y_1\oplus x_0y_0=x_1\oplus y_1\}-1\\&=2P\{x_0y_0=0\}-1\\&=\frac{1}{2}\ne 0\end{aligned}$$

Therefore, $f(X,Y)$ is not a second-order correlation immunity function.

## V. CONCLUSION

From the perspective of Boolean function, this paper deduces a recurrence formula which does not require a recall of the carry function about addition modulo $2^n$, and convert addition modulo $2^n$ into a $n$-dimensional vector Boolean function. Then the cryptographic properties of $n$-dimensional vector Boolean function of $X+Y$ mod $2^n$ is explored. The results indicate that the component functions of $X+Y$ mod $2^n$ are all $k$th-order quasi-Bent functions. It is also proved that $X+Y$ mod $2^n$ have the first-order correlation-immunity rather than the second-order correlation-immunity. The results presented in the paper can enable us to analyze the algorithm which contains addition modulo $2^n$ and XOR on computational complexity and linearity cryptanalysis method.

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